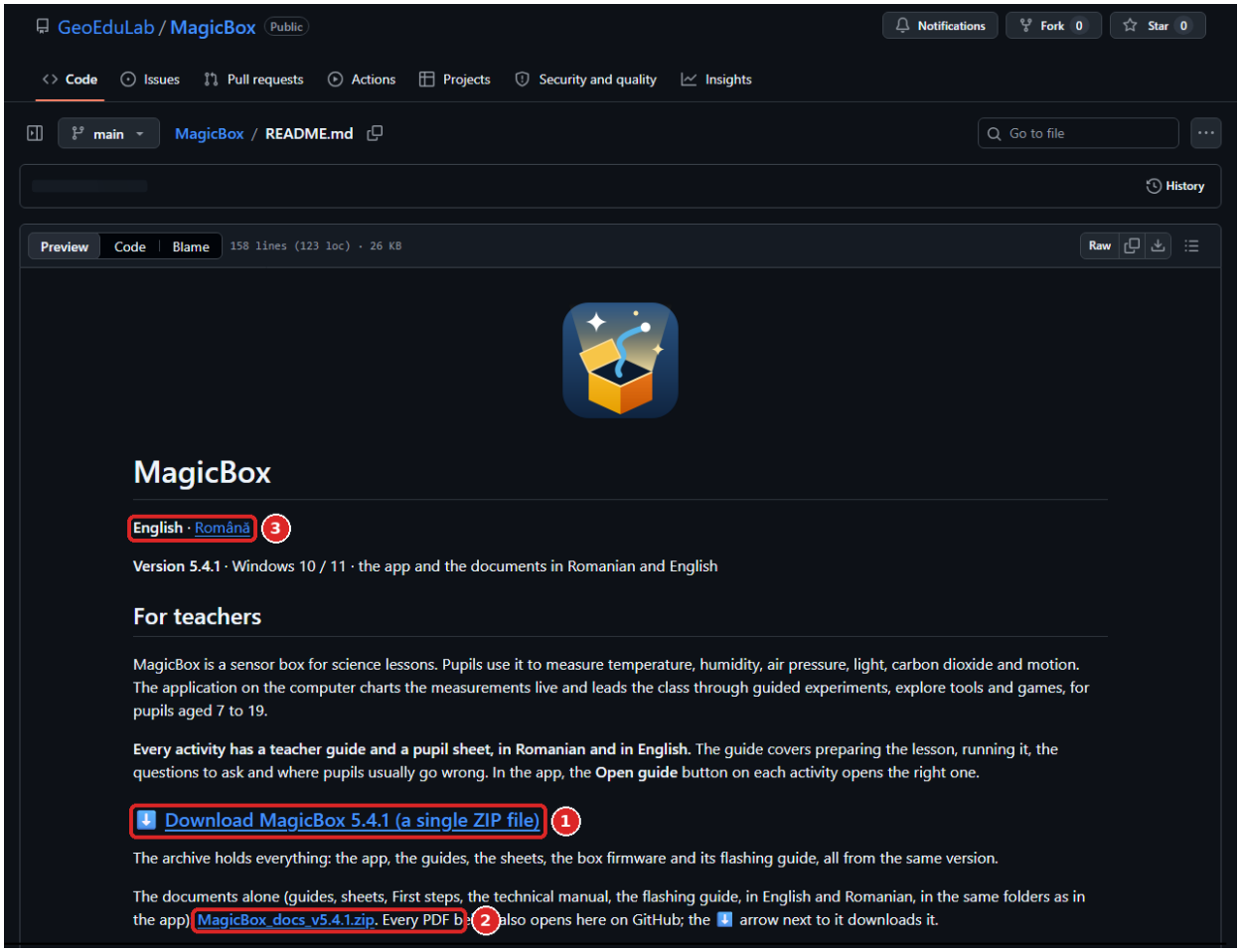
User Manual

1. Downloading the application

The application, the manuals and the box firmware are published together, on one page: `https://github.com/GeoEduLab/MagicBox`. Nobody needs a GitHub account.

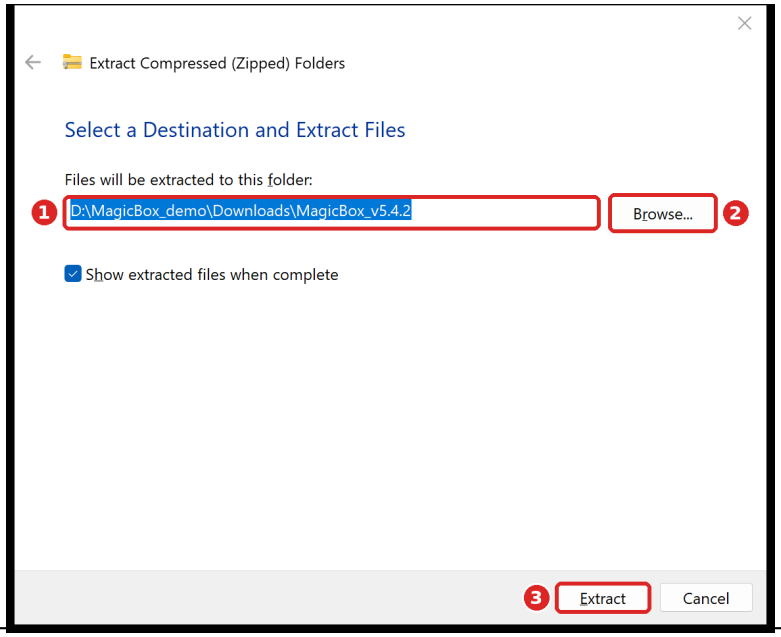


	On screen	What it does
1	Download MagicBox … (a single ZIP file)	The whole package, about 85 MB: the application, the guides and sheets in four languages, the box firmware. This is the only thing to download. Do not use the green **Code** button: it is meant for programmers.
2	MagicBox_docs_….zip	The documents alone (about 20 MB), in the same folders as in the package: for a memory stick or a shared drive. Each PDF on the page can also be read on GitHub (click its name) or saved on its own (the ↓ arrow next to it).
3	English · Română	The same page in the other language.

The version number on the page grows with every release; the latest is always at the top. What changed is in `CHANGELOG.txt` inside the package.

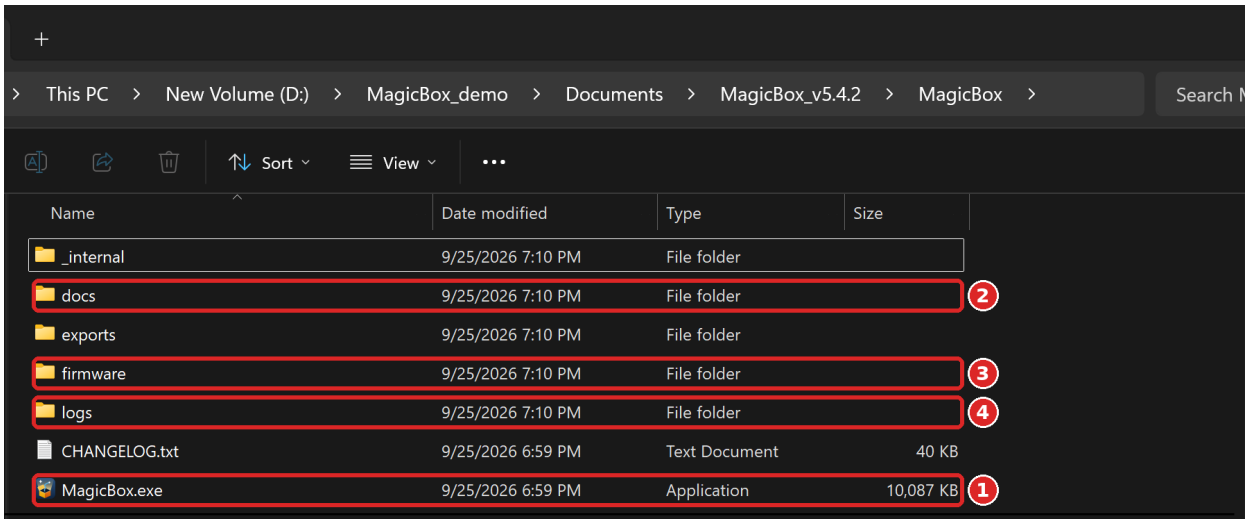
2. Installing and starting

1. Unzip. Right-click the downloaded ZIP → **Extract All...**. This window opens:



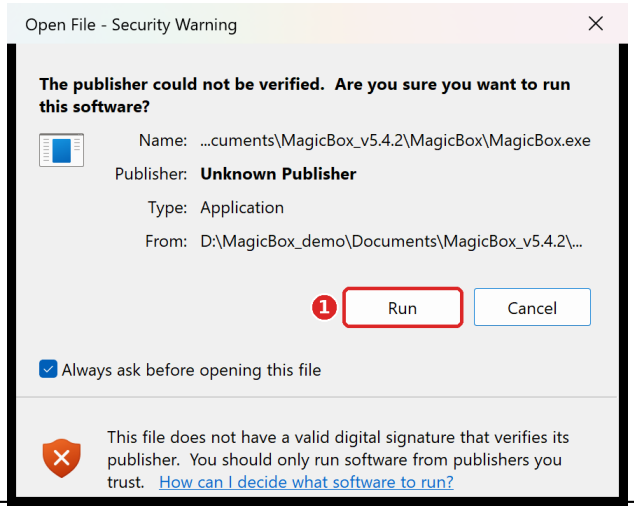
	On screen	What it does
1	Files will be extracted to this folder	Where the application goes. Choose *Documents* or a memory stick. Not *Program Files*: Windows does not let the application save its data there.
2	Browse...	Picks another folder.
3	Extract	Unzips. Never run the application from inside the ZIP.

2. Start. Open the `MagicBox` folder. Keep it whole: the application needs everything in it.



	On screen	What it does
1	MagicBox.exe	Double-click it to start the application. Nothing needs installing.
2	docs	All the manuals, `ro` and `en`.
3	firmware	The box program and its flashing guide.
4	logs	The measurements, saved by themselves. Exports go to `exports`; what changed is in `CHANGELOG.txt`.

3. The Windows warning. The application has no commercial signature, so the first time Windows asks whether to run it:



	On screen	What it does
1	Run	Starts the application. That is all the warning means.

On some computers the warning reads *"Windows protected your PC"* instead: click **More info**, then **Run anyway**.

4. The network. The first time, Windows may ask whether the application may use the network: tick **Private networks**, then **Allow access**. Without it the boxes cannot connect over WiFi.

5. The language. The application starts **in English**. For another language: **Settings** (last on the left) → **Language:**, then press **Restart now**: the application opens again in that language by itself (section 11).



	On screen	What it does
1	Language	Română, English, Français or Deutsch. The guides and manuals exist in the same four languages.

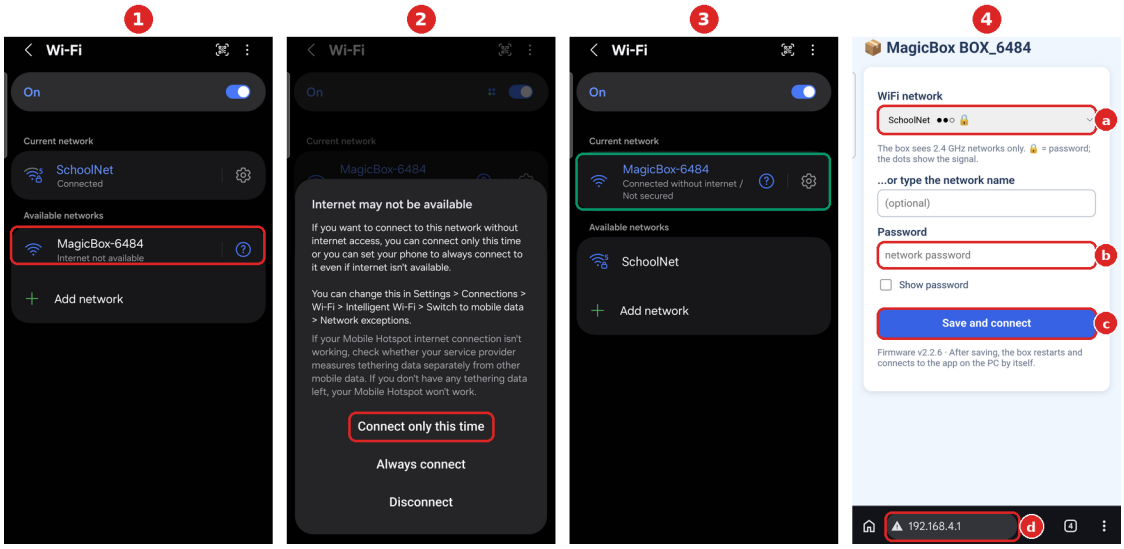
3. Connecting a box

The sensors first. A box has three sockets for four sensors (weather, light, CO_2, motion), so one is always left out. Plug in what the activity needs *with the box switched off*: the box looks for its sensors only when it starts. The activity card says which sensor it needs.

USB or WiFi, one box or many. A box connects to the computer over a USB cable or over WiFi. Several boxes can be connected to the same computer at once, over USB, over WiFi or some each way; each gets its own card on the **Home** page. A box already connected over WiFi can also be plugged into USB without any harm: the application knows it is the same box and does not count it twice.

Over USB. Use a **USB Type-C data cable** (some phone-charging cables carry no data). After a few seconds the box appears on the **Home** page. If it does not appear within 10 seconds, press **Rescan**; if it still does not appear, Windows lacks the board's driver: install **CP210x** (`https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers`) or **CH340** (`https://www.wch-ic.com/downloads/CH341SER_EXE.html`) once.

Over WiFi (once per box, from a phone):

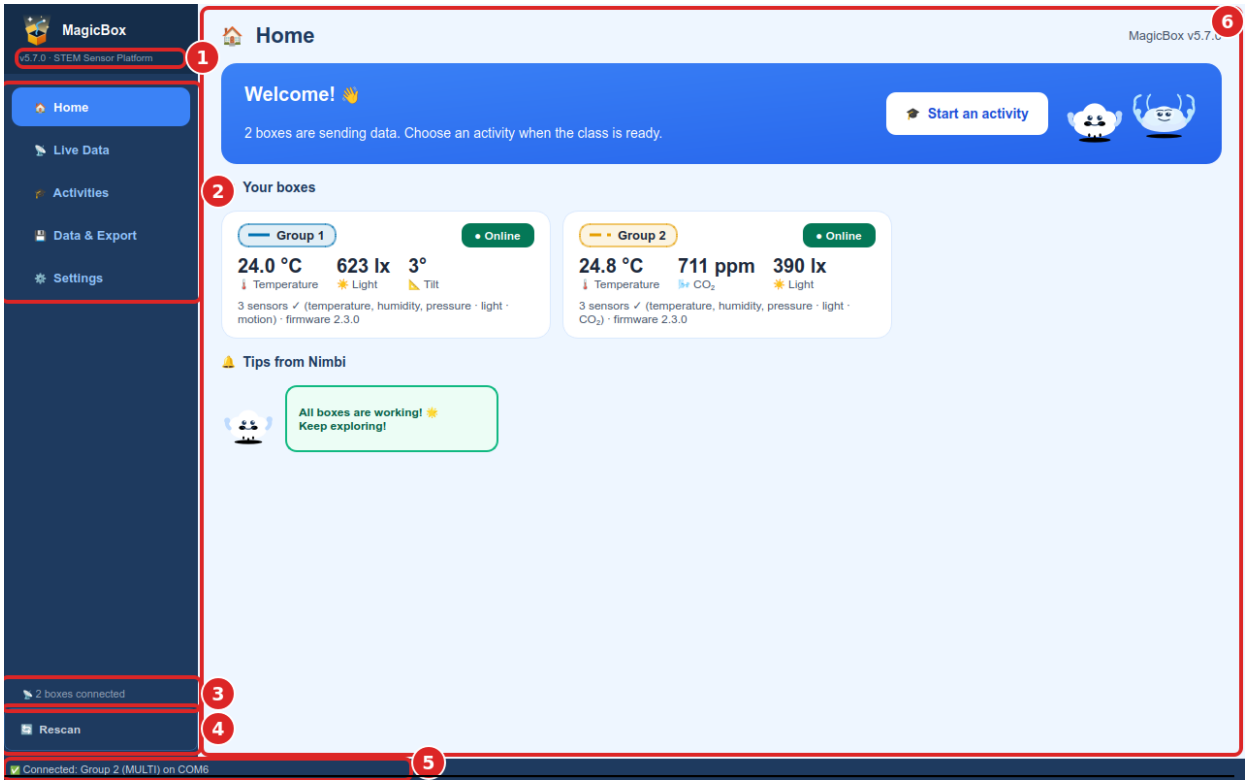


1. Switch the box on. The first time it opens a network called **MagicBox-XXXX** (XXXX = the end of the box ID). Tap it in the phone's WiFi settings.
2. The phone says the network has no internet: choose **Connect only this time**.
3. The phone stays on MagicBox-XXXX, "without internet": that is normal.
4. The setup page opens by itself; if not, open `192.168.4.1` in the browser (d). Pick the class network (a), type its password (b), tap **Save and connect** (c). The box restarts, joins the network and appears in the application by itself.

The box sees **2.4 GHz networks only**. The computer and the boxes must be on the same network; if the school network does not work (user name *and* password, or devices that cannot see each other), use a phone hotspot. To change network: hold the box's **BOOT** button while switching it on.

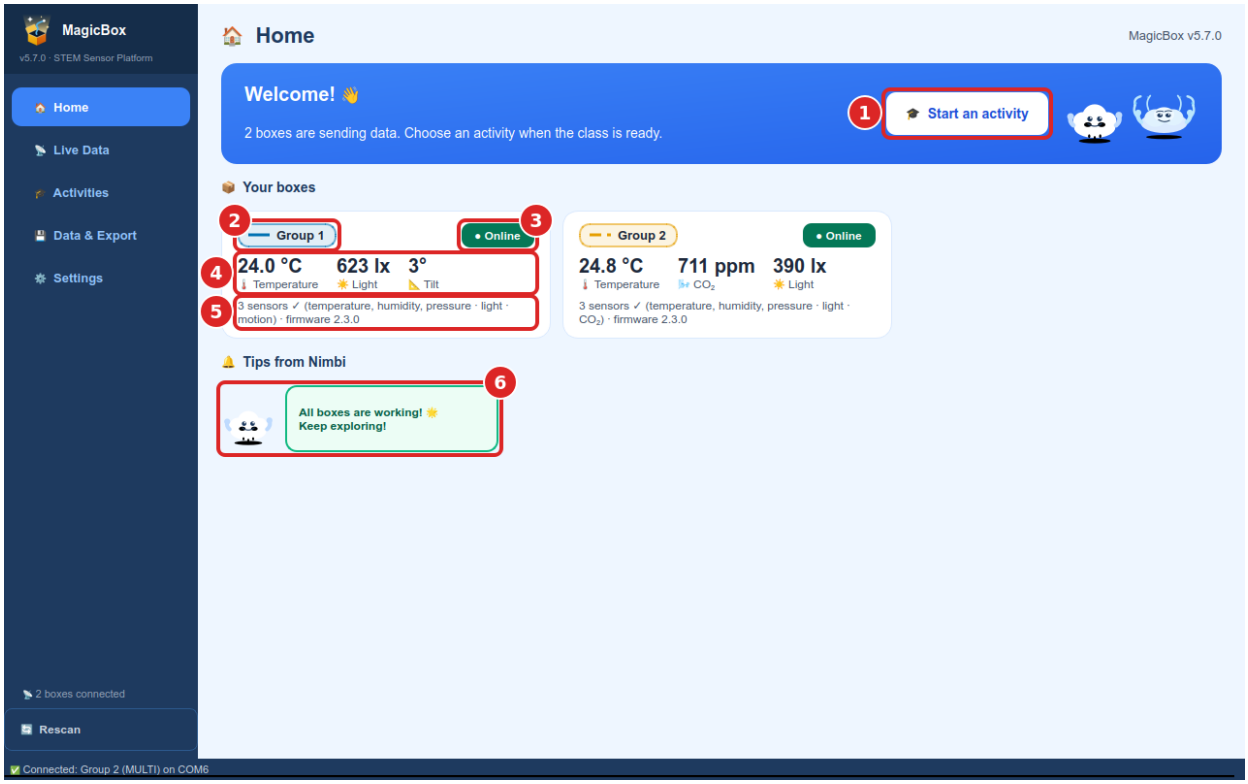
CO_2 arrives last: the CO_2 sensor needs up to a minute after power-on before its first value. The other values come at once.

4. The main window



	On screen	What it does
1	v5.7.0 · STEM Sensor Platform	The application's version. Mention it when asking for help.
2	Home, Live Data, Activities, Data & Export, Settings	The five pages. **Data & Export** appears only at the Scientist and Teacher levels (section 11).
3	2 boxes connected	How many boxes are sending data right now.
4	Rescan	Looks again for boxes on the USB ports: after plugging a box in late, or after closing the Arduino IDE. WiFi boxes are found without it.
5	Status bar	The last connection message (which box, on which port).
6	The page	The page chosen on the left.

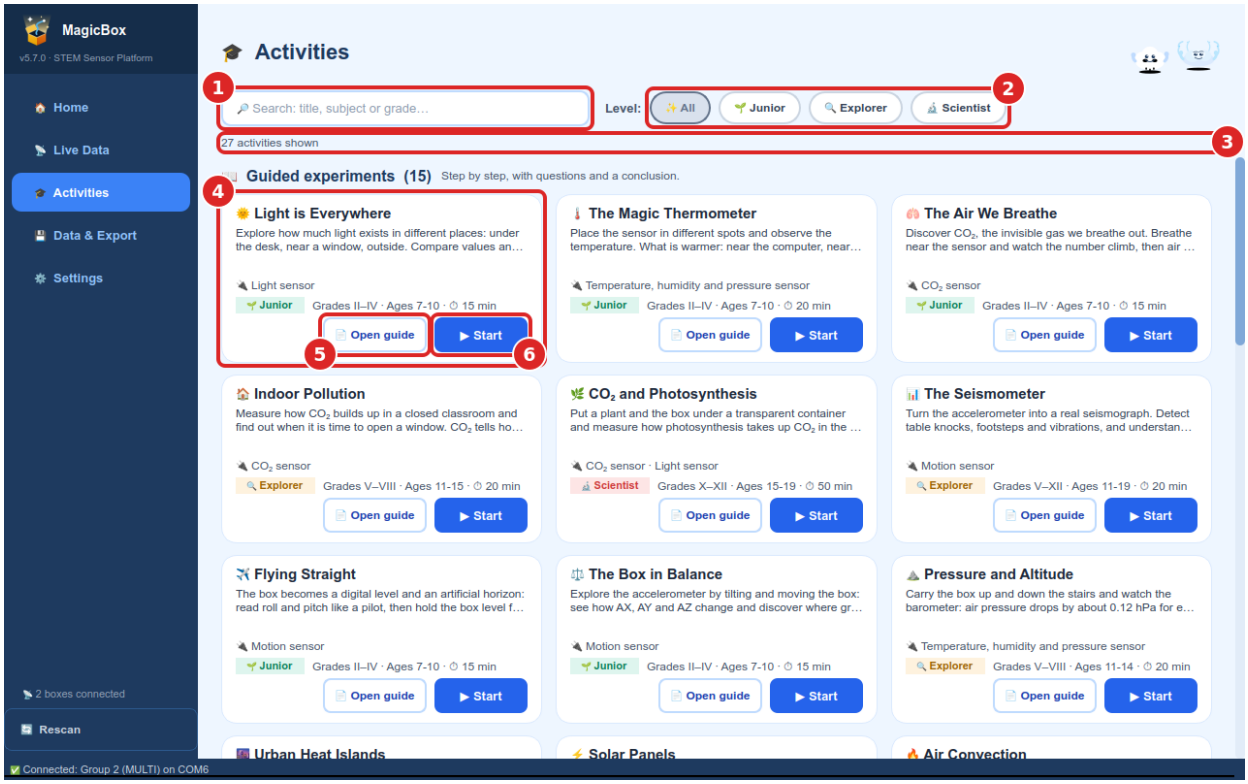
5. Home



	On screen	What it does
1	Start an activity	Goes to the **Activities** page.
2	The box's name	A name the class recognises ("Group 1"), set in **Settings** → **Box names**. The box keeps this colour and line style on every chart.
3	Online / Offline	Whether the box is sending data now. *Offline*: check the USB cable or the WiFi, then press **Rescan**.
4	The latest values	One value for each sensor plugged into the box: temperature, CO_2, light, and **Tilt** (how far the box leans from lying flat) for the motion sensor. "—" means the value has not arrived yet.
5	3 sensors … · firmware	Which sensors the box has and its firmware version.
6	Tips from Nimbi	Short advice: a box that went offline, air that needs changing, a box with old firmware.

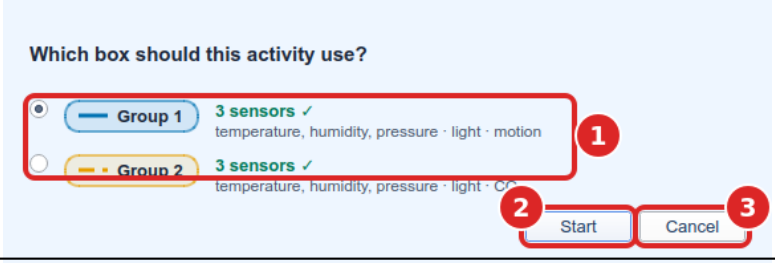
When a newer MagicBox is published, a green note appears under the welcome banner: **Download** opens the GitHub page (section 1). Download and unzip the new package as the first time; the measurements and settings stay.

6. Activities



	On screen	What it does
1	Search	Finds activities by title, subject or grade (“CO2”, “physics”, “VII”).
2	Level: All, Junior, Explorer, Scientist	Shows only the activities of one age level. Which activities the pupils see at all is set in **Settings → Visible activities**.
3	27 activities shown	How many cards the search and the filter leave.
4	An activity card	Its title and aim, **the sensor to plug in**, the level, the grades and the time it takes. The cards come in three groups: guided experiments, exploration tools, games.
5	Open guide	Opens the **teacher guide** or the **pupil sheet** (PDF) of that activity, in either language.
6	Start	Opens the activity in its own window. If no box has the sensor it needs, the application says which sensor to plug in.

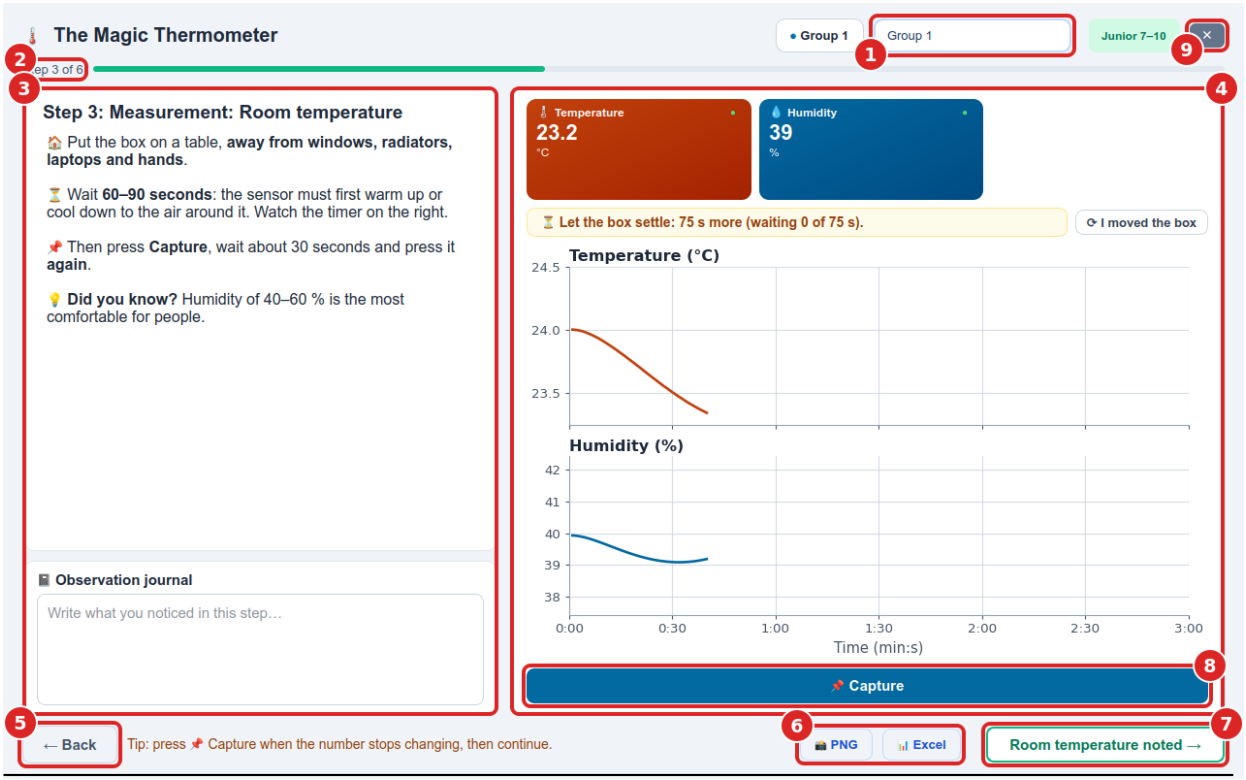
Choosing the box. When several boxes could run the activity, the application asks which one:



	On screen	What it does
1	The boxes	Each box with its name and the sensors it has. Pick one.
2	Start	Opens the activity on the chosen box.
3	Cancel	Back to the activities, nothing opened.

7. A guided experiment

Guided experiments lead the class step by step: an instruction, a measurement, a question, a conclusion. They all look like this (here: The Magic Thermometer).



	On screen	What it does
1	The box	The box whose data the step uses. With more than one box, choose another here.
2	Step 3 of 6	Where the class is; the bar fills as it advances.
3	The instruction	What to do in this step, and below it the **Observation journal**, where pupils write what they notice (it goes into the Excel file).
4	Values and charts	The live measurement: the current value, a countdown while the sensor settles (**I moved the box** starts it again), and the chart from the start of the step.
5	Back	The previous step.
6	PNG, Excel	Save this step: the charts as one picture (PNG), or the measurements and the journal as an Excel file.
7	Continue	The next step; its label says what the class has done (here "Room temperature noted →"). On the last step it reads **Finish** and shows the results, with a button that opens the folder of saved files.
8	Capture	Keeps the value shown now as the result of the step. Press it when the number stops changing.
9	×	Closes the experiment. The measurements stay saved in `logs`.

8. An exploration tool or a game

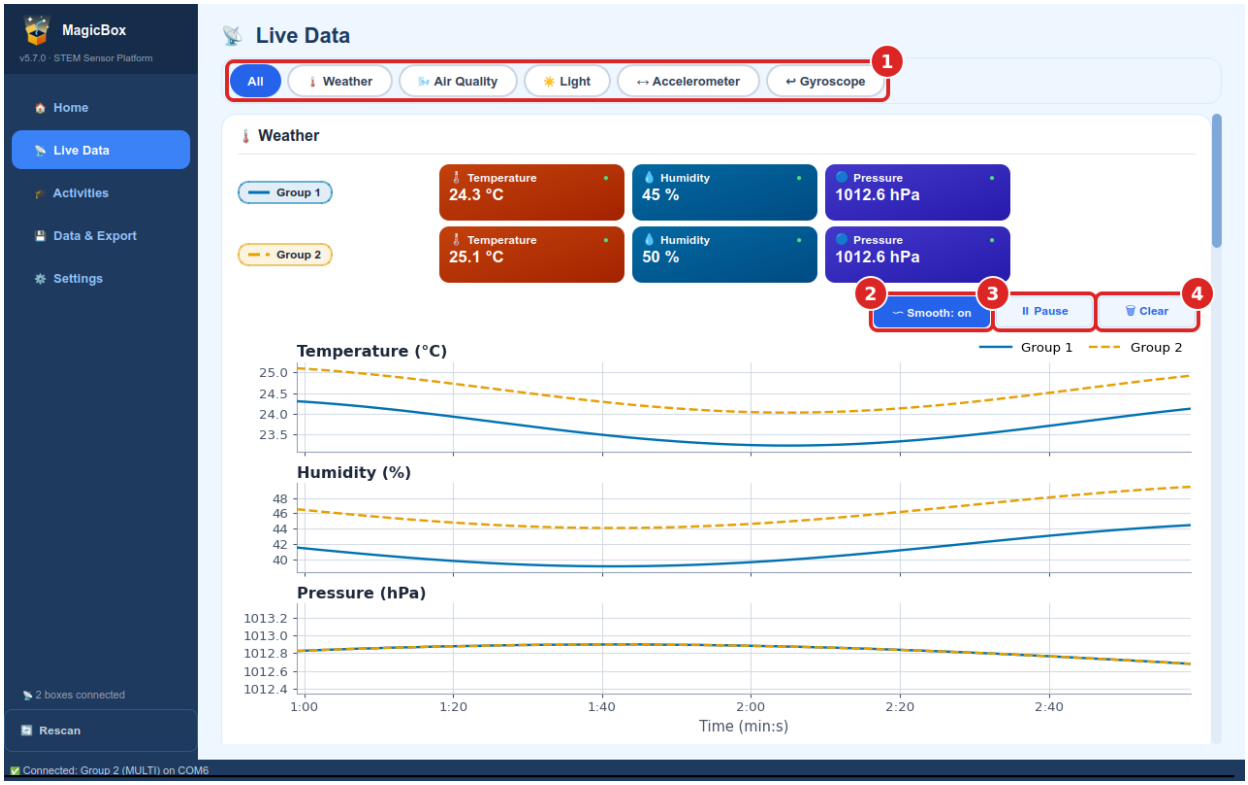
Tools (Weather Station, Air Quality, Seismology…) and games (Among the Clouds, Earthquake!…) have no steps: they show the data their own way. The top of the window is the same (here: the Weather Station).



	On screen	What it does
1	Change box	Shows another box's data.
2	Close	Closes the window.
3	Export data	Saves the measurements of this session as Excel or CSV: the raw readings (not the smoothed line), with the time of each one.
4	Tabs	**Live charts**, **Analysis** (the numbers worked out from the data) and **Theory** (the science behind it).
5	Smooth, Pause, Clear	See section 9.
6	Value cards	The current value of each quantity.

9. Live Data

All the boxes at once, grouped by what they measure.

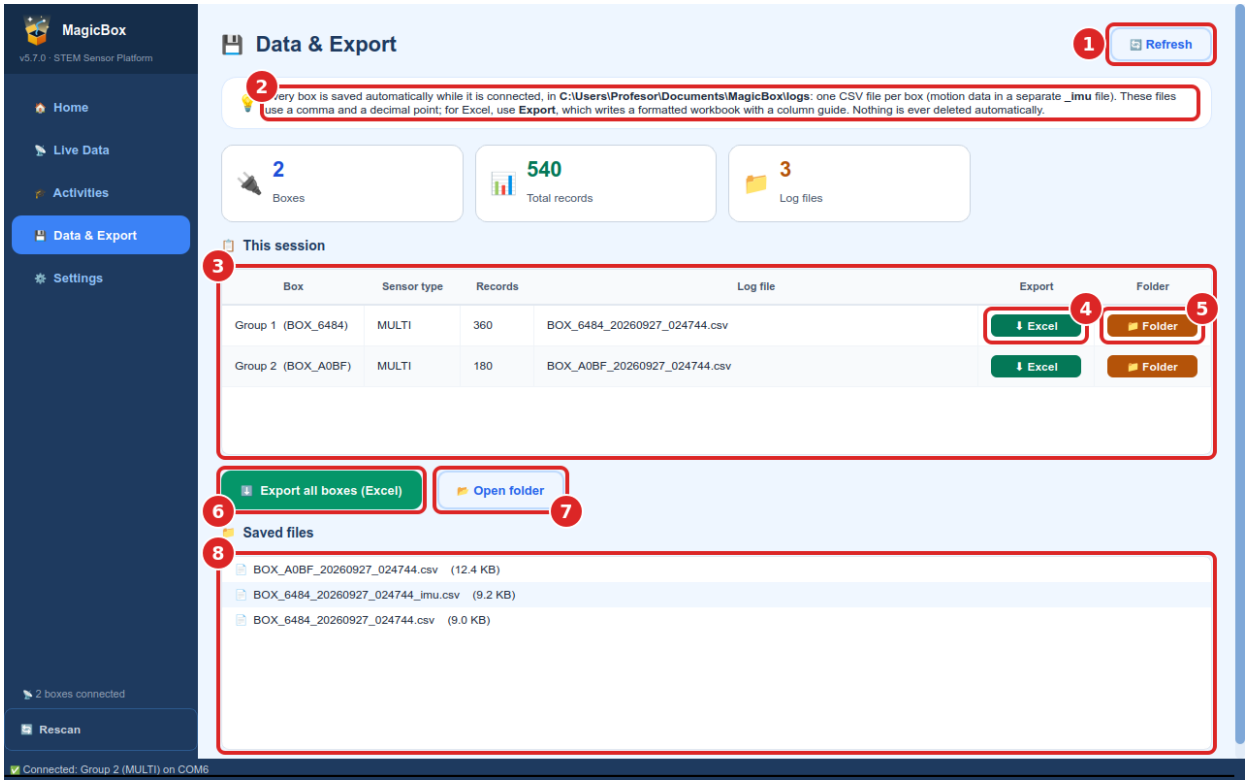


	On screen	What it does
1	All, Weather, Air quality, Light, Accelerometer, Gyroscope	Shows every group, or only one.
2	Smooth: on / off	Smoothing hides the sensor's small jitter. Turn it off to see every raw reading. The saved data are always the raw ones.
3	Pause / Resume	Freezes the chart so the class can look at it. The box keeps measuring and the data keep being saved.
4	Clear	Starts the chart again from 0:00. Nothing saved is deleted.

Each box has its own colour and line style, the same on every screen; the legend names them.

10. Data & Export

What the application has saved, and exporting it for Excel. Visible at the Scientist and Teacher levels.

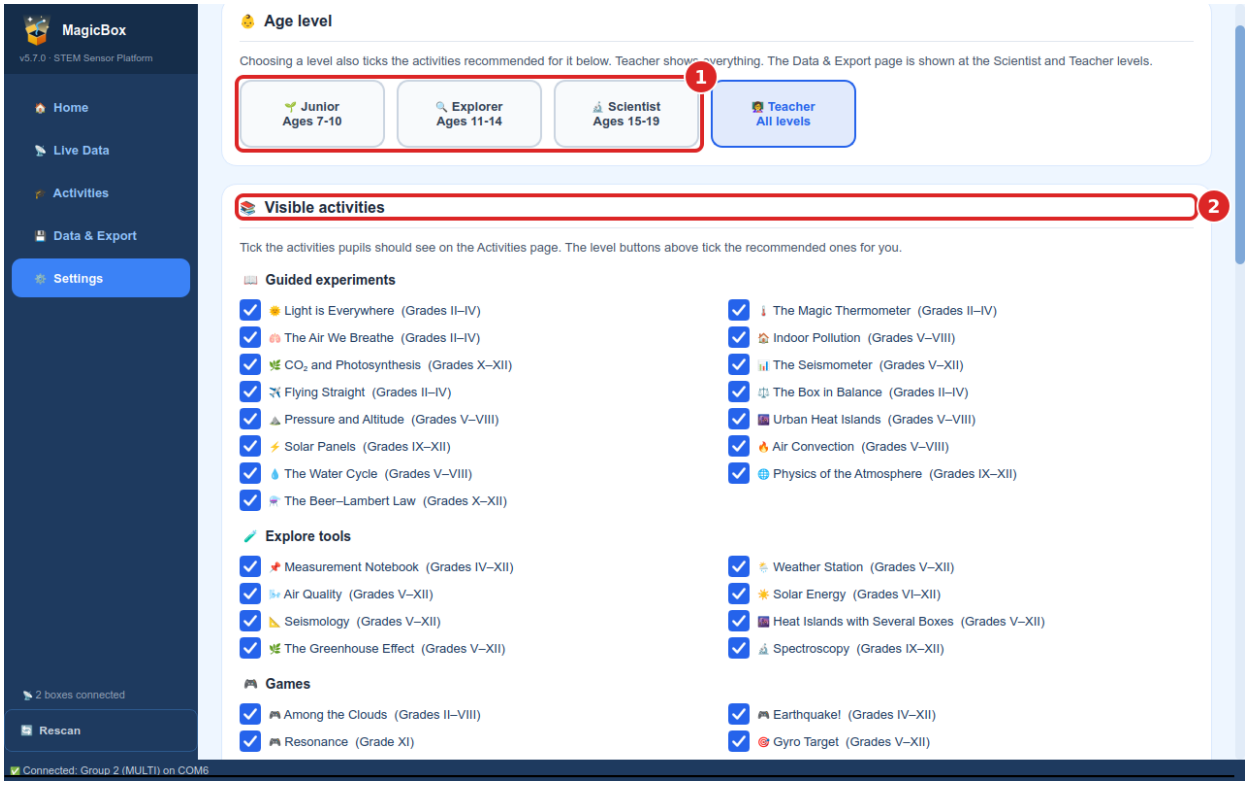


	On screen	What it does
1	Refresh	Reads the list of files again.
2	Where the data go	Every box's data are saved by themselves, while it is connected, in the `logs` folder next to the application: one CSV file per box (motion in a separate `_imu` file).
3	Current session	Each box: its name, how many records, its log file.
4	Excel	Exports that box as a formatted Excel workbook, with a sheet explaining the columns.
5	Folder	Opens the folder with that box's file.
6	Export all boxes (Excel)	Asks for a place and writes a lesson folder with one workbook per box.
7	Open folder	Opens the `logs` folder.
8	Saved files	All the log files; double-click one to open it.

The CSV files use a comma and a decimal point, as every program reads them; for Excel in Romanian use **Excel** or **Export all boxes**, which write a proper workbook. Nothing is deleted without asking (section 11).

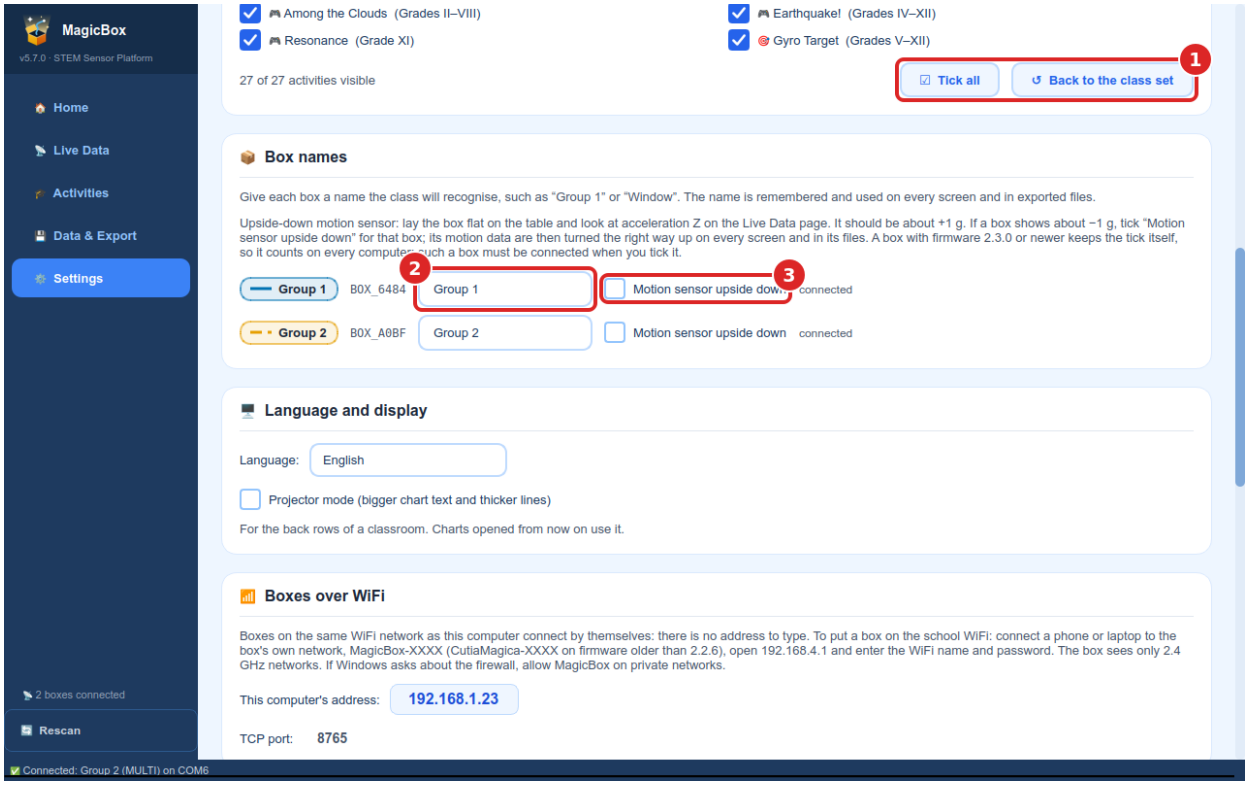
11. Settings

Age level and visible activities



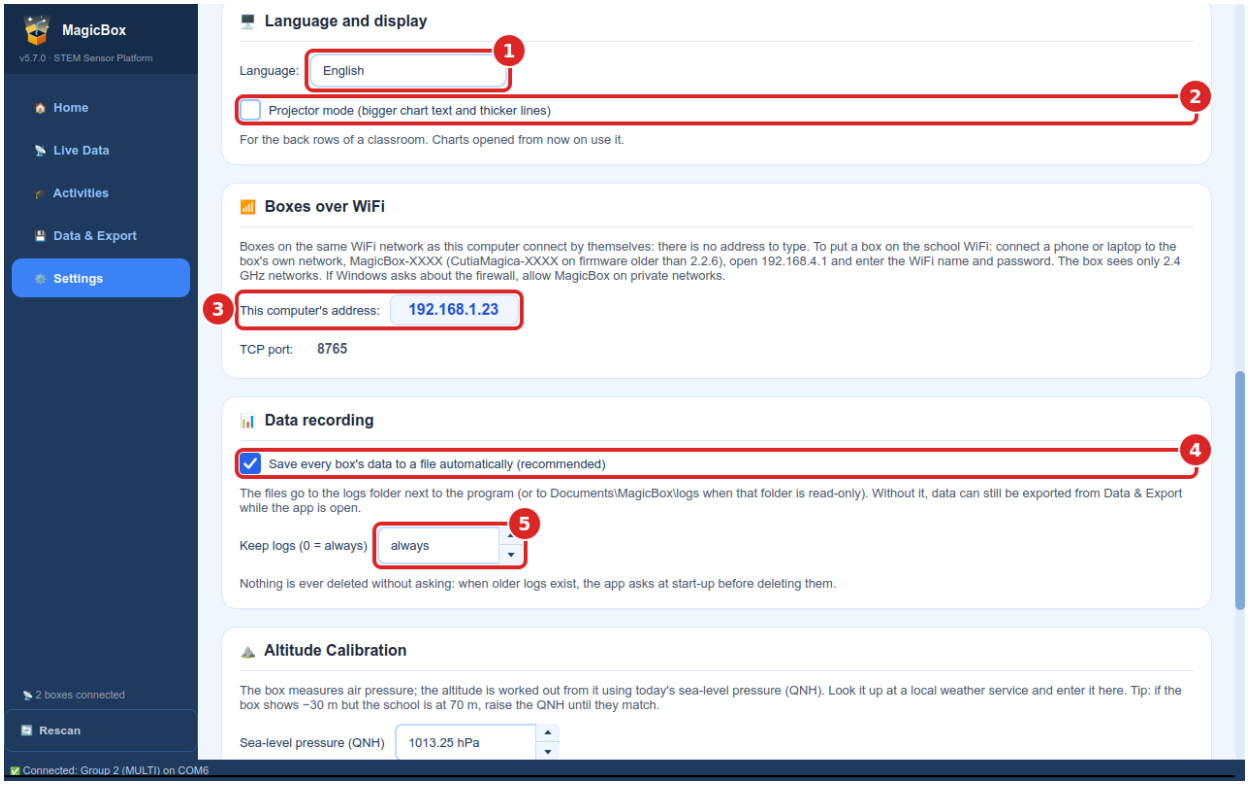
	On screen	What it does
1	Junior, Explorer, Scientist, Teacher	The age of the class. A level ticks the activities recommended for it and sets how much the screens show (Junior hides the advanced chart buttons). **Teacher** shows everything.
2	Visible activities	One tick per activity: only the ticked ones appear on the Activities page.

Box names



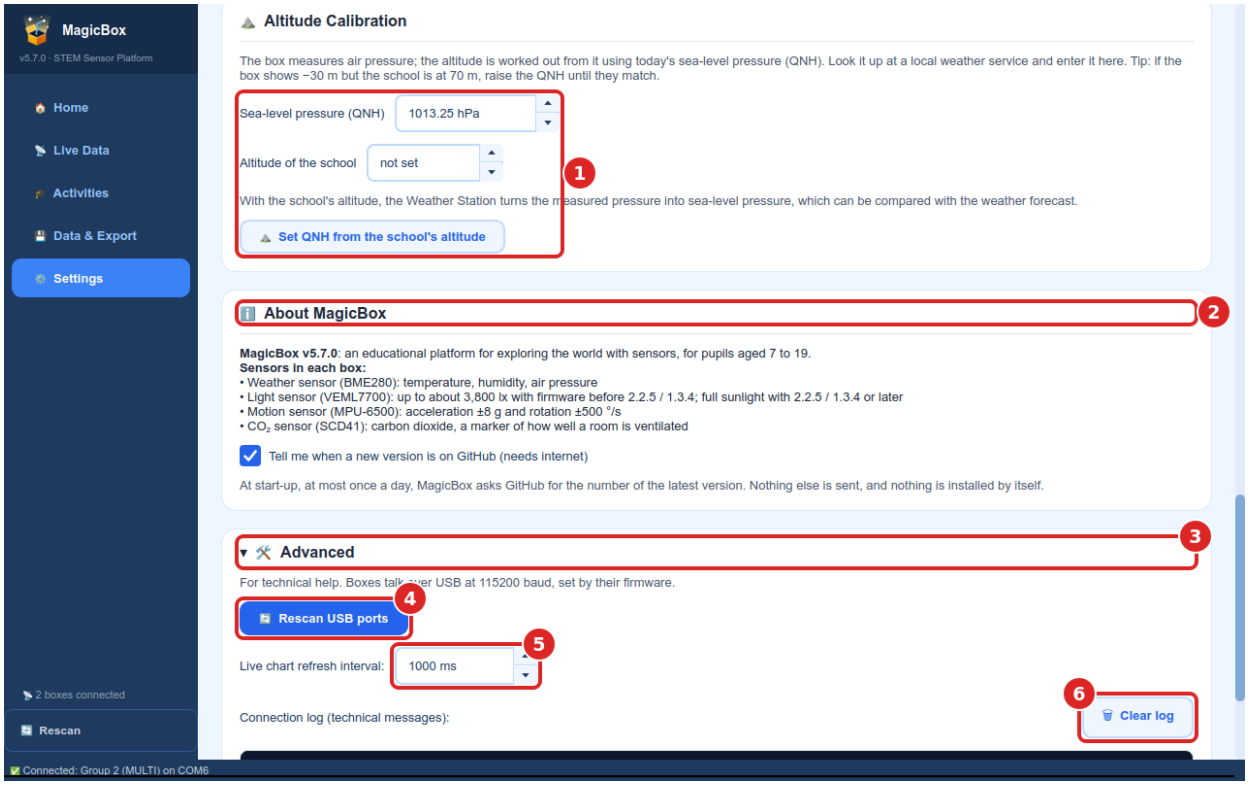
	On screen	What it does
1	Tick all / Back to the class set	Ticks every activity, or only the ones recommended for the chosen level.
2	The name	The name the class sees on every screen and in the exported files (“Group 1”, “Window”).
3	Motion sensor upside down	For a box whose motion sensor is mounted the other way up: lying flat, it shows acceleration Z of about −1 g instead of +1 g. Tick it once for that box; every screen and file is then right. A box with firmware 2.3.0 or newer keeps the tick itself, so it counts on every computer: it must be connected when you tick it, and its row then says “motion sensor setting kept in the box”.

Language, WiFi and data recording



	On screen	What it does
1	Language	Română, English, Français or Deutsch. A **Restart now** button appears: the application opens again in the new language, the boxes reconnect by themselves.
2	Projector mode	Larger text and thicker lines on the charts, for the back rows.
3	This computer's address	For technical help only: WiFi boxes find the computer by themselves.
4	Save every box's data to a file	On by default. Off: nothing goes to `logs`, but **Data & Export** can still export while the application is open.
5	Keep logs	After how many days old log files may be deleted; *always* keeps them. The application always asks before deleting.

Altitude, about, advanced



	On screen	What it does
1	Sea-level pressure, altitude of the school	The box works out altitude from air pressure. Enter today's sea-level pressure (QNH) from a weather service and the school's altitude; the Weather Station then shows the pressure a forecast would. With the altitude entered and a box connected, **Set QNH from the school's altitude** works the QNH out from the pressure the box reads now, so the box shows the school's altitude.
2	About MagicBox	The version, the sensors in each box, and **Tell me when a new version is on GitHub**: on by default; at start-up, at most once a day, the application asks GitHub for the latest version number. Nothing else is sent; without internet nothing happens.
3	Advanced	Opens the technical settings.
4	Rescan USB ports	Same as **Rescan** on the left.
5	Live chart refresh interval	How often the charts redraw. Raise it on a slow computer.
6	Clear log	Empties the connection log below, the first place to look when a box will not connect ("Sensors it reports: …").

12. When something does not work

What you see	What to do
The box does not appear over USB	A data cable; the CP210x/CH340 driver; close the Arduino IDE; **Rescan**.
The box does not appear over WiFi	Computer and box on the same 2.4 GHz network; the firewall permission (Private networks); a phone hotspot.
A sensor is missing from the box's card	Switch the box off, push the sensor firmly into its socket, switch it on.
CO_2 shows "—"	Wait up to a minute after switching the box on.
The activity asks for a sensor	The box lacks it: switch off, plug it in, switch on.
Lying flat, acceleration Z is about -1 g	Tick **Motion sensor upside down** for that box.
The tick springs back: "Not changed: the box did not answer."	The box (firmware 2.3.0 or newer) keeps the tick itself: connect it and tick again.
The Excel export is not saved	The file is open in Excel: close it.
The application should be in another language	**Settings** → **Language:**, then **Restart now**.

More: *First steps* (`docs\en\first_steps.pdf`, two pages) and the *Technical manual* (`docs\en\technical_manual.pdf`). Loading the firmware onto a box: `firmware\FLASHING_GUIDE.pdf`.